

Bassam Abdulrahim

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SUMMARY

AI engineer specializing in generative AI systems: retrieval, agentic workflows, and document understanding. First hire at Entropy, now engineering lead of a 6-person team, hands-on technical owner of Cortex, an enterprise knowledge management platform I designed, built, and continue to develop, and builder of AI solutions for enterprise clients.

EXPERIENCE

Entropy

Riyadh, Saudi Arabia

Data and AI arm of Saudi Azm (a publicly listed technology company). Founding engineer and first hire; grew with the company to 50+ employees.

Engineering Lead

Jun 2026 to Present

- Lead a 6-person engineering team while remaining hands-on in the codebase: set architectural direction and engineering conventions, interview and onboard technical hires, and mentor data scientists and software engineers on system design and LLM application development.
- Build AI solutions for enterprise clients, owning technical design and implementation from scoping through delivery, across agentic AI, document understanding, information retrieval, and computer vision, including collecting data and fine-tuning YOLO object detection models.
- Continue as technical owner of Cortex, driving its architecture and building its core systems alongside the team.

Senior Data Scientist

Jan 2025 to Jun 2026

- Built AI solutions and proofs of concept for enterprise clients in parallel with Cortex, across agentic AI, information retrieval, and document understanding.
- Served as the technical point of contact in client engagements: ran discovery sessions to scope requirements, presented solution architectures and live demos to client stakeholders, and wrote the technical sections of proposals.
- Built and shipped Cortex (cortex.entropy.sa), an enterprise AI knowledge management platform, as lead engineer. Chose the full stack, designed the architecture, and took it from concept to a deployed product.
- Built Arabic-first document ingestion combining OCR and vision-language models to handle scanned and mixed-layout Arabic PDFs, where conventional text extraction pipelines fail. This was the hardest and highest-impact problem in the product.
- Designed automated knowledge graph construction over ingested documents: entity and relationship extraction, cross-document linking, and graph persistence in Neo4j. Built for knowledge exploration, surfacing connections that span multiple documents, rather than as a retrieval path for question answering.
- Built agentic search on Pydantic AI over hybrid retrieval, with query transformation and a pluggable reranking layer supporting multiple backends (self-hosted open-source models and the Cohere reranker), plus automatic organization of ingested documents (titling, summarization, tagging, classification, inter-document linking).
- Built programmatic generation of Word documents and PowerPoint decks from retrieved and synthesized content, with layout, structure, and formatting generated end to end rather than templated.
- Adopted Restate for durable execution of long-running agentic and ingestion workflows, making multi-step pipelines resumable and fault-tolerant rather than dependent on ad-hoc retry logic.
- Established the testing approach: unit tests alongside an end-to-end harness that provisions a disposable replica of the environment and its data, exercises the system through its public API, records results, then tears the replica down. Both run in CI through GitHub Actions.
- Own model selection for Cortex and many client solutions, choosing separately for each role in the pipeline: open-weight LLMs for generation (Qwen, Command R, Kimi, GPT-OSS), OCR and vision-language models for extraction (Chandra OCR, Qwen-VL), and embedding and reranking models for retrieval, all under self-hosted deployment constraints.
- Cortex stack: Python/FastAPI, Pydantic AI, TypeScript/React/Next.js/Tailwind, PostgreSQL, Qdrant, Neo4j, Restate, Keycloak, ClamAV, MinIO, Nginx, Docker. Instrumented for production with OpenTelemetry, Prometheus, Grafana, and Loki covering metrics, tracing, and log aggregation. Built entirely on self-hosted open-source models with no dependency on third-party AI APIs. Deployed on AWS (EC2, CloudFront, Amplify) for internal use and client trials, and portable to on-premise deployment with no vendor lock-in, a requirement for enterprise clients with data residency constraints.
- Designed and built two internal libraries for document extraction and retrieval, underpinning the company's AI work. Both have pluggable provider backends and evaluation designed in from the start rather than bolted on, and are now reused across Cortex, client proofs of concept, and other delivered solutions:
 - **ContentMiner**: parsing, OCR, vision-language extraction, pre- and post-processing, and evaluation harnesses for measuring extraction quality across document types.
 - **RetrievalEngine**: semantic retrieval, BM25, hybrid retrieval, vector database abstractions, reranking, and retrieval evaluation.
- Built a compliance review agent that cross-checks documents against regulatory articles clause by clause, flagging

conflicts and gaps with citations back to source regulation. Demoed successfully to prospective clients.

Data Scientist

Sep 2023 to Jan 2025

- Built and delivered 15+ AI prototypes for enterprise clients, mostly focused on generative AI across document processing, retrieval, and analysis. Typically built on open-source models to meet on-premise and data residency constraints.
- Scoped problems directly with enterprise clients, presented solutions to their stakeholders, and wrote the technical content for company proposals.
- Co-built Faheim, a generative-AI financial analyst for the Saudi stock market that reached beta release with real users, built over a knowledge base of thousands of financial statements and annual reports. Worked across data collection and preparation, LLM system development with LangChain, FastAPI backend development, and deployment to AWS via Docker, ECR, and ECS.

Master Works

Riyadh, Saudi Arabia

Software Engineering Trainee

Jun 2023 to Sep 2023

- Built backend services in Laravel with Docker-based local environments, and deployed containerized services to Kubernetes using kubectl.

PROJECTS

Hidaya: Open-source Android application for Muslim users, **18,000+ installs** github.com/BassamAlim/Hidaya
Built and maintained solo over 1,000+ commits. Migrated the live app from Java/XML to Kotlin and Jetpack Compose without disrupting existing users. Feature-sliced architecture holding one consistent pattern across 37 screens, with 15 repositories over Room and DataStore, offline astronomical prayer-time calculation, alarm-based notification scheduling with boot recovery, home-screen widgets, and full Arabic/English RTL support. Kotlin, Compose, Hilt, Room, Firebase, GitHub Actions.

AirAware: ML system for predicting flight delays in Saudi Arabia (*published, see below*)

Co-built an end-to-end pipeline covering multi-source data scraping with Selenium, feature engineering, and comparative evaluation of XGBoost, CatBoost, LightGBM, and MLP models. Python, NumPy, pandas, scikit-learn, matplotlib, MySQL.

Asaar: Serverless price tracking across online retailers

Go/Gin service deployed as an AWS Lambda function, scraping and tracking product prices across multiple online stores, with a native Android client for browsing price history.

PUBLICATION

M. Alfarhood, R. Alotaibi, **B. Abdulrahim**, A. Einieh, M. Almousa, and A. Alkhanifer. "Predicting Flight Delays with Machine Learning: A Case Study from Saudi Arabian Airlines." *International Journal of Aerospace Engineering*, vol. 2024, Article ID 3385463, Wiley, 2024. doi.org/10.1155/2024/3385463

EDUCATION

B.Sc. Computer Science, King Saud University

2019 to 2023

First-class honors

CERTIFICATIONS & COMMUNITY

- Applied Data Science with Python Specialization, University of Michigan (Coursera), 2022
- Machine Learning Specialization, DeepLearning.AI, 2023
- Mentored participating teams at the Transformation Hackathon, King Saud University, 2025

SKILLS

Programming: Python, Kotlin, Java, Go, TypeScript/JavaScript, SQL

AI/ML: Agentic AI development, RAG, hybrid retrieval (semantic + BM25), reranking, retrieval and extraction evaluation, knowledge graphs, OCR/VLM document understanding, computer vision (YOLO, Ultralytics), LangChain, Pydantic AI, PyTorch, scikit-learn, NumPy, pandas

Backend & Infrastructure: FastAPI, Restate, Docker, Kubernetes, AWS (EC2, Lambda, ECS, ECR, CloudFront, Amplify), Nginx, Keycloak, MinIO, OpenTelemetry, Prometheus, Grafana, Loki, Laravel, Gin

Data Stores: PostgreSQL, MySQL, Qdrant, Neo4j

Frontend: React, Next.js, Tailwind, Jetpack Compose

Tools: GitHub Actions, Selenium, Playwright, Git

Languages: Arabic (native), English (fluent)